MSHA Narrative Extraction Comparison

Final Manual, Online, Programmatic, and Entity Analysis

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Shared topic: CLASSIFICATION = HANDLING OF MATERIALS
Document 1: DOCUMENT_NO=220021510007; MINE_ID=1202010
Document 2: DOCUMENT_NO=220122850018; MINE_ID=1202295

Purpose:
Compare manually identified terms and entities with online and Python extraction methods.
Evaluate useful-term coverage, noise, phrase preservation, preprocessing effects,
entity extraction, and normalization of equivalent surface forms.

Important scoring note:
The online tools returned ordered lists without numerical scores or counts.
Their output is recorded as rank; literal occurrence counts are calculated separately.

Executive Summary

Two coal-mine narratives classified as HANDLING OF MATERIALS were compared. Both involve roof-bolting materials, retaining bands or straps, unexpected movement of metal, and a struck-by injury mechanism. The narratives use different wording and describe different immediate objects: a strap in Document 1 and a roof bolt in Document 2.

The manual ground truth emphasizes complete safety concepts rather than every frequent word. Online lists are evaluated using rank and exact ground-truth coverage because the services did not provide comparable scores, counts, or reliable entity types.

FiveFilters returned a mix of useful phrases and generic fragments. Wordcount.com preserved longer event and equipment phrases and therefore provides more context. Dandelion returned a small set of short object labels. This demonstrates that a general entity extractor may have limited coverage in short technical incident narratives, where the most useful entities are equipment, materials, body parts, hazards, work activities, and event mechanisms.

The report includes exact-match precision, recall, and F1, but those metrics are conservative: semantically similar phrases receive no exact-match credit. The equivalence and broader-concept tables are therefore needed to interpret related forms such as bolts/roof bolts and band/strap/bundle straps.

Selected Narrative 1: Complete Stored Text

DOCUMENT 1

Record identifier: DOCUMENT_NO=220021510007; MINE_ID=1202010
Shared topic: CLASSIFICATION = HANDLING OF MATERIALS
ACCIDENT_DT: 2002-05-22 00:00:00
CLASSIFICATION: HANDLING OF MATERIALS
ACCIDENT_TYPE: Struck by... NEC
ACTIVITY: Handling supplies or material
INJURY_SOURCE: METAL,NEC(PIPE,WIRE,NAIL)
NATURE_INJURY: CUT,LACER,PUNCT-OPN WOUND
INJ_BODY_PART: EYE(S) OPTIC NERVE/VISON
DEGREE_INJURY: DAYS AWAY FROM WORK ONLY
OCCUPATION: Apprentice, Trainee
COAL_METAL_IND: C

FULL NARRATIVE TEXT

EE WAS LOADING SUPPLIES FOR THE INTAKE ROOF BOLTER.T HERE WAS ABOUT 30 FOUR FOOT ROOF BOLTS LEFT ONT
HE BOLTER ALONG WITH THE BUNDLE STRAPS OR BANDS THAT HELD THE BOLTS. EE WAS REMOVING THESE BOLTS SO
HE COULD PLACE A NEW BUNDLE ON THE BOLTER. AS HE WAS PULLING THESE BOLTS OFF BY HAND HE CAUGHT ONE
OF THESE STRAPS LAYING LOOSE ON TOP OF THE BOLTER AND THE TEND OF THE STRAP STRUCK H

Character count: 384
Word count: 77
Note: this is the complete stored database value; a 384-character record may still be source-truncated.

Selected Narrative 2: Complete Stored Text

DOCUMENT 2

Record identifier: DOCUMENT_NO=220122850018; MINE_ID=1202295
Shared topic: CLASSIFICATION = HANDLING OF MATERIALS
ACCIDENT_DT: 2012-09-25 00:00:00
CLASSIFICATION: HANDLING OF MATERIALS
ACCIDENT_TYPE: Struck by flying object
ACTIVITY: Handling supplies or material
INJURY_SOURCE: ROOF (ROCK) BOLTS
NATURE_INJURY: FRACTURE,CHIP
INJ_BODY_PART: MOUTH/LIP/TEETH/TONGUE/THROAT/TASTE
DEGREE_INJURY: DAYS AWAY FROM WORK ONLY
OCCUPATION: Laborer, Bull gang, Parts runner, Roustabout, Roof trimmer/scaler
COAL_METAL_IND: C

FULL NARRATIVE TEXT

The IE was preparing to load roof bolts into the supply tray for the roof bolter. When he cut the bands loose, one of the roof bolts came up and struck him in the mouth loosening the bridge in his mouth. It was determined that the IE had set the edge of the scoop bucket down on one of the roof bolts. He had a permanent tooth pulled on 10/4/12, that was loosened during the incident.

Character count: 384
Word count: 76
Note: this is the complete stored database value; a 384-character record may still be source-truncated.

Step 1 - Estimated Manual Ground Truth

term	kind	entity_type	canonical	doc1_count	doc2_count
roof bolter	ENTITY	EQUIPMENT	roof-bolting machine	1	1
roof bolts	ENTITY	MATERIAL	roof bolt	1	3
loading supplies	DOMAIN_CONCEPT	ACTIVITY	material handling	1	0
intake roof bolter	ENTITY	EQUIPMENT	roof-bolting machine	1	0
four foot roof bolts	ENTITY	MATERIAL	roof bolt	1	0
bundle straps	ENTITY	MATERIAL_RESTRAINT	retaining band	1	0
bands	ENTITY	MATERIAL_RESTRAINT	retaining band	1	1
removing these bolts	TERM	ACTIVITY	remove roof bolts	1	0
new bundle	ENTITY	MATERIAL	roof-bolt bundle	1	0
pulling these bolts off by hand	DOMAIN_CONCEPT	ACTIVITY	manual material handling	1	0
straps laying loose	DOMAIN_CONCEPT	HAZARD	loose retaining band	1	0
caught one of these straps	DOMAIN_CONCEPT	TRIGGER	strap caught during handling	1	0
strap	ENTITY	MATERIAL_RESTRAINT	retaining band	1	0
EE	ENTITY	PERSON_REFERENCE	employee	2	0
load roof bolts	TERM	ACTIVITY	load roof bolts	0	1
supply tray	ENTITY	EQUIPMENT_COMPONENT	roof-bolter supply tray	0	1
scoop bucket	ENTITY	EQUIPMENT_COMPONENT	scoop bucket	0	1
cut the bands loose	DOMAIN_CONCEPT	TRIGGER	release retaining bands	0	1
roof bolt came up	DOMAIN_CONCEPT	HAZARD_EVENT	unexpected roof-bolt movement	0	0
struck him in the mouth	DOMAIN_CONCEPT	MECHANISM	struck-by event	0	1

Step 1 - Estimated Manual Ground Truth (continued)

term	kind	entity_type	canonical	doc1_count	doc2_count
mouth	ENTITY	BODY_PART	mouth	0	2
bridge	ENTITY	DENTAL_APPLIANCE	dental bridge	0	1
permanent tooth	ENTITY	BODY_PART	tooth	0	1
IE	ENTITY	PERSON_REFERENCE	injured employee	0	2

Step 1 - Ground-Truth Rationale

The manual ground truth is a reasoned estimate based on close reading of both narratives. It includes equipment, materials, material restraints, work activities, hazards, event triggers, injury mechanisms, person references, and body parts. Generic connecting words were excluded even when an automated extractor returned them.

The document counts are literal whole-phrase counts. A zero does not necessarily mean the concept is absent; the narrative may express it with different wording. Exact phrase counts are intentionally kept separate from online rank and Python relevance scores.

Consolidated Ground Truth and Engine Comparison

manual_term_or_e	kind	canonical	doc1_count	doc2_count	Count n-grams D1	Count n-grams D2	TF-IDF D1	TF-IDF D2
roof bolter	ENTITY	roof-bolting machine	1	1				
roof bolts	ENTITY	roof bolt	1	3		3.6		0.1518
loading supplies	DOMAIN_CONCEPT	material handling	1	0				
intake roof bolter	ENTITY	roof-bolting machine	1	0				
four foot roof bolts	ENTITY	roof bolt	1	0				
bundle straps	ENTITY	retaining band	1	0				
bands	ENTITY	retaining band	1	1				
removing these bolts	TERM	remove roof bolts	1	0				
new bundle	ENTITY	roof-bolt bundle	1	0				
pulling these bolts off by hand	DOMAIN_CONCEPT	manual material handling	1	0				
straps laying loose	DOMAIN_CONCEPT	loose retaining band	1	0			0.095	
caught one of these straps	DOMAIN_CONCEPT	strap caught during handling	1	0				
strap	ENTITY	retaining band	1	0			0.095	
EE	ENTITY	employee	2	0	2.0		0.1608	
load roof bolts	TERM	load roof bolts	0	1				
supply tray	ENTITY	roof-bolter supply tray	0	1				0.1016
scoop bucket	ENTITY	scoop bucket	0	1				
cut the bands loose	DOMAIN_CONCEPT	release retaining bands	0	1				

Consolidated Ground Truth and Engine Comparison (continued)

manual_term_or_e	kind	canonical	doc1_count	doc2_count	Count n-grams D1	Count n-grams D2	TF-IDF D1	TF-IDF D2
roof bolt came up	DOMAIN_CONCEPT	unexpected roof-bolt movement	0	0				
struck him in the mouth	DOMAIN_CONCEPT	struck-by event	0	1				
mouth	ENTITY	mouth	0	2		2.0		0.1721
bridge	ENTITY	dental bridge	0	1				
permanent tooth	ENTITY	tooth	0	1				
IE	ENTITY	injured employee	0	2		2.0		0.1721

Consolidated Ground Truth and Engine Comparison

manual_term_or_e	kind	canonical	doc1_count	doc2_count	RAKE-like D1	RAKE-like D2	NLTK FreqDist D1	NLTK FreqDist D2
roof bolter	ENTITY	roof-bolting machine	1	1	6.25	4.5		
roof bolts	ENTITY	roof bolt	1	3	6.75	15.5		
loading supplies	DOMAIN_CONCEPT	material handling	1	0				
intake roof bolter	ENTITY	roof-bolting machine	1	0	9.25			
four foot roof bolts	ENTITY	roof bolt	1	0				
bundle straps	ENTITY	retaining band	1	0				
bands	ENTITY	retaining band	1	1		3.0	1	1
removing these bolts	TERM	remove roof bolts	1	0				
new bundle	ENTITY	roof-bolt bundle	1	0				
pulling these bolts off by hand	DOMAIN_CONCEPT	manual material handling	1	0				
straps laying loose	DOMAIN_CONCEPT	loose retaining band	1	0	8.5			
caught one of these straps	DOMAIN_CONCEPT	strap caught during handling	1	0				
strap	ENTITY	retaining band	1	0				
EE	ENTITY	employee	2	0				
load roof bolts	TERM	load roof bolts	0	1		8.1667		
supply tray	ENTITY	roof-bolter supply tray	0	1		4.0		
scoop bucket	ENTITY	scoop bucket	0	1		4.0		
cut the bands loose	DOMAIN_CONCEPT	release retaining bands	0	1				

Consolidated Ground Truth and Engine Comparison (continued)

manual_term_or_e	kind	canonical	doc1_count	doc2_count	RAKE-like D1	RAKE-like D2	NLTK FreqDist D1	NLTK FreqDist D2
roof bolt came up	DOMAIN_CONCEPT	unexpected roof-bolt movement	0	0				
struck him in the mouth	DOMAIN_CONCEPT	struck-by event	0	1				
mouth	ENTITY	mouth	0	2				2
bridge	ENTITY	dental bridge	0	1				1
permanent tooth	ENTITY	tooth	0	1		6.0		
IE	ENTITY	injured employee	0	2				

Consolidated Ground Truth and Engine Comparison

manual_term_or_e	kind	canonical	doc1_count	doc2_count	FiveFilters D1	FiveFilters D2	Wordcount.com D	Wordcount.com D2
roof bolter	ENTITY	roof-bolting machine	1	1		Rank 4		Rank 3
roof bolts	ENTITY	roof bolt	1	3		Rank 2		Rank 1
loading supplies	DOMAIN_CONCEPT	material handling	1	0	Rank 2		Rank 2	
intake roof bolter	ENTITY	roof-bolting machine	1	0			Rank 1	
four foot roof bolts	ENTITY	roof bolt	1	0			Rank 3	
bundle straps	ENTITY	retaining band	1	0	Rank 4		Rank 4	
bands	ENTITY	retaining band	1	1	Rank 9	Rank 8		
removing these bolts	TERM	remove roof bolts	1	0	Rank 1			
new bundle	ENTITY	roof-bolt bundle	1	0	Rank 6			
pulling these bolts off by hand	DOMAIN_CONCEPT	manual material handling	1	0				
straps laying loose	DOMAIN_CONCEPT	loose retaining band	1	0				
caught one of these straps	DOMAIN_CONCEPT	strap caught during handling	1	0				
strap	ENTITY	retaining band	1	0	Rank 15			
EE	ENTITY	employee	2	0	Rank 7			
load roof bolts	TERM	load roof bolts	0	1		Rank 1		
supply tray	ENTITY	roof-bolter supply tray	0	1		Rank 3		Rank 2
scoop bucket	ENTITY	scoop bucket	0	1		Rank 5		
cut the bands loose	DOMAIN_CONCEPT	release retaining bands	0	1				Rank 5

Consolidated Ground Truth and Engine Comparison (continued)

manual_term_or_e	kind	canonical	doc1_count	doc2_count	FiveFilters D1	FiveFilters D2	Wordcount.com D	Wordcount.com D2
roof bolt came up	DOMAIN_CONCEPT	unexpected roof-bolt movement	0	0				
struck him in the mouth	DOMAIN_CONCEPT	struck-by event	0	1				
mouth	ENTITY	mouth	0	2		Rank 7		
bridge	ENTITY	dental bridge	0	1		Rank 9		
permanent tooth	ENTITY	tooth	0	1				Rank 11
IE	ENTITY	injured employee	0	2		Rank 6		

Consolidated Ground Truth and Engine Comparison

manual_term_or_e	kind	canonical	doc1_count	doc2_count	Dandelion D1	Dandelion D2	Wordcount phrase	Wordcount phrase/en
roof bolter	ENTITY	roof-bolting machine	1	1				Rank 3
roof bolts	ENTITY	roof bolt	1	3				Rank 1
loading supplies	DOMAIN_CONCEPT	material handling	1	0				
intake roof bolter	ENTITY	roof-bolting machine	1	0			Rank 1	
four foot roof bolts	ENTITY	roof bolt	1	0			Rank 2	
bundle straps	ENTITY	retaining band	1	0			Rank 3	
bands	ENTITY	retaining band	1	1				
removing these bolts	TERM	remove roof bolts	1	0				
new bundle	ENTITY	roof-bolt bundle	1	0				
pulling these bolts off by hand	DOMAIN_CONCEPT	manual material handling	1	0				
straps laying loose	DOMAIN_CONCEPT	loose retaining band	1	0				
caught one of these straps	DOMAIN_CONCEPT	strap caught during handling	1	0				
strap	ENTITY	retaining band	1	0	Rank 4			
EE	ENTITY	employee	2	0	Rank 1			
load roof bolts	TERM	load roof bolts	0	1				
supply tray	ENTITY	roof-bolter supply tray	0	1				Rank 2
scoop bucket	ENTITY	scoop bucket	0	1				
cut the bands loose	DOMAIN_CONCEPT	release retaining bands	0	1				

Consolidated Ground Truth and Engine Comparison (continued)

manual_term_or_e	kind	canonical	doc1_count	doc2_count	Dandelion D1	Dandelion D2	Wordcount phrase	Wordcount phrase/en
roof bolt came up	DOMAIN_CONCEPT	unexpected roof-bolt movement	0	0				
struck him in the mouth	DOMAIN_CONCEPT	struck-by event	0	1				
mouth	ENTITY	mouth	0	2				
bridge	ENTITY	dental bridge	0	1				
permanent tooth	ENTITY	tooth	0	1		Rank 6		Rank 7
IE	ENTITY	injured employee	0	2				

Consolidated Ground Truth and Engine Comparison

manual_term_or_entity	kind	canonical	doc1_count	doc2_count	Domain dictionary D1	Domain dictionary D2
roof bolter	ENTITY	roof-bolting machine	1	1	roof-bolting work	roof-bolting work
roof bolts	ENTITY	roof bolt	1	3	roof-bolting work	roof-bolting work
loading supplies	DOMAIN_CONCEPT	material handling	1	0	material handling	
intake roof bolter	ENTITY	roof-bolting machine	1	0	roof-bolting work	
four foot roof bolts	ENTITY	roof bolt	1	0		
bundle straps	ENTITY	retaining band	1	0	material restraint	
bands	ENTITY	retaining band	1	1	material restraint	material restraint
removing these bolts	TERM	remove roof bolts	1	0		
new bundle	ENTITY	roof-bolt bundle	1	0		
pulling these bolts off by hand	DOMAIN_CONCEPT	manual material handling	1	0		
straps laying loose	DOMAIN_CONCEPT	loose retaining band	1	0		
caught one of these straps	DOMAIN_CONCEPT	strap caught during handling	1	0		
strap	ENTITY	retaining band	1	0	material restraint	
EE	ENTITY	employee	2	0	person reference	
load roof bolts	TERM	load roof bolts	0	1		material handling
supply tray	ENTITY	roof-bolter supply tray	0	1		roof-bolter components
scoop bucket	ENTITY	scoop bucket	0	1		mobile equipment components
cut the bands loose	DOMAIN_CONCEPT	release retaining bands	0	1		

Consolidated Ground Truth and Engine Comparison (continued)

manual_term_or_entity	kind	canonical	doc1_count	doc2_count	Domain dictionary D1	Domain dictionary D2
roof bolt came up	DOMAIN_CONCEPT	unexpected roof-bolt movement	0	0		
struck him in the mouth	DOMAIN_CONCEPT	struck-by event	0	1		unexpected material movement
mouth	ENTITY	mouth	0	2		oral injury
bridge	ENTITY	dental bridge	0	1		oral injury
permanent tooth	ENTITY	tooth	0	1		oral injury
IE	ENTITY	injured employee	0	2		person reference

Step 2 - FiveFilters - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	FiveFilters	removing these bolts	Rank 1	1
2	FiveFilters	loading supplies	Rank 2	1
3	FiveFilters	bolter along	Rank 3	1
4	FiveFilters	bundle straps	Rank 4	1
5	FiveFilters	could place	Rank 5	1
6	FiveFilters	new bundle	Rank 6	1
7	FiveFilters	ee	Rank 7	2
8	FiveFilters	bolter	Rank 8	4
9	FiveFilters	bands	Rank 9	1
10	FiveFilters	bolts	Rank 10	4
11	FiveFilters	as	Rank 11	1
12	FiveFilters	caught	Rank 12	1
13	FiveFilters	top	Rank 13	1
14	FiveFilters	tend	Rank 14	1
15	FiveFilters	strap	Rank 15	1

Step 2 - FiveFilters - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	FiveFilters	load roof bolts	Rank 1	1
2	FiveFilters	roof bolts	Rank 2	3
3	FiveFilters	supply tray	Rank 3	1
4	FiveFilters	roof bolter	Rank 4	1
5	FiveFilters	scoop bucket	Rank 5	1
6	FiveFilters	ie	Rank 6	2
7	FiveFilters	mouth	Rank 7	2
8	FiveFilters	bands	Rank 8	1
9	FiveFilters	bridge	Rank 9	1
10	FiveFilters	edge	Rank 10	1
11	FiveFilters	tooth	Rank 11	1
12	FiveFilters	incident	Rank 12	1

Step 2 - Wordcount.com - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	Wordcount.com	intake roof bolter	Rank 1	1
2	Wordcount.com	loading supplies	Rank 2	1
3	Wordcount.com	four foot roof bolts	Rank 3	1
4	Wordcount.com	bundle straps	Rank 4	1
5	Wordcount.com	bolts on the bolter	Rank 5	0
6	Wordcount.com	removing bolts	Rank 6	0
7	Wordcount.com	placing a new bundle	Rank 7	0
8	Wordcount.com	pulling bolts off by hand	Rank 8	0
9	Wordcount.com	strap laying loose	Rank 9	0
10	Wordcount.com	strap tension	Rank 10	0
11	Wordcount.com	tend of the strap struck	Rank 11	1
12	Wordcount.com	caught one strap	Rank 12	0
13	Wordcount.com	ee was removing bolts	Rank 13	0

Step 2 - Wordcount.com - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	Wordcount.com	roof bolts	Rank 1	3
2	Wordcount.com	supply tray	Rank 2	1
3	Wordcount.com	roof bolter	Rank 3	1
4	Wordcount.com	ie (incident employee)	Rank 4	0
5	Wordcount.com	cut the bands loose	Rank 5	1
6	Wordcount.com	roof bolt struck him in the mouth	Rank 6	0
7	Wordcount.com	bridge in his mouth loosened	Rank 7	0
8	Wordcount.com	edge of the scoop bucket	Rank 8	1
9	Wordcount.com	set the edge of the scoop bucket down	Rank 9	1
10	Wordcount.com	permanent tooth pulled	Rank 10	1
11	Wordcount.com	permanent tooth	Rank 11	1
12	Wordcount.com	incident injury	Rank 12	0
13	Wordcount.com	loosened tooth during incident	Rank 13	0
14	Wordcount.com	10/4/12	Rank 14	1

Python Count n-grams - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	Count n-grams	bolter	4.0	4
2	Count n-grams	bolts	4.0	4
3	Count n-grams	bundle	2.0	2
4	Count n-grams	ee	2.0	2
5	Count n-grams	roof	2.0	2
6	Count n-grams	straps	2.0	2
7	Count n-grams	along bundle straps	1.4	1
8	Count n-grams	bands held bolts	1.4	1
9	Count n-grams	bolter along bundle	1.4	1
10	Count n-grams	bolter four foot	1.4	1
11	Count n-grams	bolter pulling bolts	1.4	1
12	Count n-grams	bolter tend strap	1.4	1
13	Count n-grams	bolts ee removing	1.4	1
14	Count n-grams	bolts hand caught	1.4	1
15	Count n-grams	bolts left ont	1.4	1
16	Count n-grams	bolts place new	1.4	1
17	Count n-grams	bundle bolter pulling	1.4	1
18	Count n-grams	bundle straps bands	1.4	1
19	Count n-grams	caught one straps	1.4	1
20	Count n-grams	ee loading supplies	1.4	1

Python Count n-grams - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	Count n-grams	roof	4.0	4
2	Count n-grams	roof bolts	3.6	3
3	Count n-grams	bolts	3.0	3
4	Count n-grams	one roof bolts	2.8	2
5	Count n-grams	one roof	2.4	2
6	Count n-grams	ie	2.0	2
7	Count n-grams	mouth	2.0	2
8	Count n-grams	one	2.0	2
9	Count n-grams	bands loose one	1.4	1
10	Count n-grams	bolter cut bands	1.4	1
11	Count n-grams	bolts came struck	1.4	1
12	Count n-grams	bolts permanent tooth	1.4	1
13	Count n-grams	bolts supply tray	1.4	1
14	Count n-grams	bridge mouth determined	1.4	1
15	Count n-grams	bucket one roof	1.4	1
16	Count n-grams	came struck mouth	1.4	1
17	Count n-grams	cut bands loose	1.4	1
18	Count n-grams	determined ie set	1.4	1
19	Count n-grams	edge scoop bucket	1.4	1
20	Count n-grams	ie preparing load	1.4	1

Python TF-IDF - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	TF-IDF	bolter	0.1613	4
2	TF-IDF	bolts	0.1613	4
3	TF-IDF	ee	0.1608	2
4	TF-IDF	bundle	0.1608	2
5	TF-IDF	straps	0.1608	2
6	TF-IDF	roof	0.1144	2
7	TF-IDF	straps laying loose	0.095	1
8	TF-IDF	supplies	0.095	1
9	TF-IDF	supplies intake	0.095	0
10	TF-IDF	supplies intake roof	0.095	0
11	TF-IDF	removing	0.095	1
12	TF-IDF	pulling	0.095	1
13	TF-IDF	pulling bolts	0.095	0
14	TF-IDF	pulling bolts hand	0.095	0
15	TF-IDF	straps bands	0.095	0
16	TF-IDF	roof bolts left	0.095	1
17	TF-IDF	strap	0.095	1
18	TF-IDF	strap struck	0.095	1
19	TF-IDF	top	0.095	1
20	TF-IDF	tend	0.095	1

Python TF-IDF - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	TF-IDF	roof	0.1726	4
2	TF-IDF	one roof	0.1721	0
3	TF-IDF	ie	0.1721	2
4	TF-IDF	mouth	0.1721	2
5	TF-IDF	one roof bolts	0.1721	0
6	TF-IDF	roof bolts	0.1518	3
7	TF-IDF	bolts	0.1518	3
8	TF-IDF	one	0.1224	2
9	TF-IDF	supply tray roof	0.1016	0
10	TF-IDF	struck mouth	0.1016	0
11	TF-IDF	supply	0.1016	1
12	TF-IDF	preparing load roof	0.1016	0
13	TF-IDF	preparing load	0.1016	0
14	TF-IDF	roof bolts came	0.1016	1
15	TF-IDF	roof bolts supply	0.1016	0
16	TF-IDF	roof bolts permanent	0.1016	0
17	TF-IDF	struck mouth loosening	0.1016	0
18	TF-IDF	supply tray	0.1016	1
19	TF-IDF	tray	0.1016	1
20	TF-IDF	tooth	0.1016	1

Term Extraction Metrics against Manual Ground Truth

method	returned	ground_truth_hits	precision	recall	f1
Count n-grams D1	20	0	0.000	0.000	0.000
Count n-grams D2	20	0	0.000	0.000	0.000
TF-IDF D1	20	1	0.050	0.111	0.069
TF-IDF D2	20	0	0.000	0.000	0.000
RAKE-like D1	20	1	0.050	0.111	0.069
RAKE-like D2	20	1	0.050	0.111	0.069
NLTK FreqDist D1	20	0	0.000	0.000	0.000
NLTK FreqDist D2	20	0	0.000	0.000	0.000
FiveFilters D1	15	2	0.133	0.222	0.167
FiveFilters D2	12	1	0.083	0.111	0.095
Wordcount.com D1	13	1	0.077	0.111	0.091
Wordcount.com D2	14	1	0.071	0.111	0.087

Step 3 - NLTK and Programmatic Methods

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Document 1 tokenizer: regex fallback; NLTK unavailable
Document 2 tokenizer: regex fallback; NLTK unavailable
NLTK FreqDist provides directly interpretable token counts. RAKE-like extraction ranks
candidate phrases using word frequency and co-occurrence degree. TF-IDF emphasizes terms
that distinguish one document from the other. Their numerical outputs are not the same scale.
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Step 3 - RAKE-like Phrases - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	RAKE-like	four foot roof	16.5	1
2	RAKE-like	bolts left ont	14.25	1
3	RAKE-like	foot roof bolts	12.75	1
4	RAKE-like	roof bolts left	12.75	1
5	RAKE-like	four foot	12.0	1
6	RAKE-like	left ont	12.0	1
7	RAKE-like	foot roof	10.5	1
8	RAKE-like	intake roof bolter	9.25	1
9	RAKE-like	roof	9.0	2
10	RAKE-like	bolts	9.0	4
11	RAKE-like	place new bundle	8.5	1
12	RAKE-like	straps laying loose	8.5	1
13	RAKE-like	bolts left	8.25	1
14	RAKE-like	intake roof	7.5	1
15	RAKE-like	bolter	7.0	4
16	RAKE-like	roof bolts	6.75	1
17	RAKE-like	roof bolter	6.25	1
18	RAKE-like	place new	6.0	1
19	RAKE-like	laying loose	6.0	1
20	RAKE-like	four	6.0	1

Step 3 - RAKE-like Phrases - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	RAKE-like	roof bolts	15.5	3
2	RAKE-like	roof	10.0	4
3	RAKE-like	permanent tooth pulled	9.0	1
4	RAKE-like	load roof bolts	8.1667	1
5	RAKE-like	roof bolts came	8.1667	1
6	RAKE-like	bands loose one	8.0	1
7	RAKE-like	bolts	8.0	3
8	RAKE-like	bands loose	6.0	1
9	RAKE-like	permanent tooth	6.0	1
10	RAKE-like	tooth pulled	6.0	1
11	RAKE-like	bolts came	5.6667	1
12	RAKE-like	load roof	5.5	1
13	RAKE-like	loose one	5.0	1
14	RAKE-like	roof bolter	4.5	1
15	RAKE-like	supply tray	4.0	1
16	RAKE-like	scoop bucket	4.0	1
17	RAKE-like	one	4.0	2
18	RAKE-like	mouth loosening	3.5	1
19	RAKE-like	load	3.0	1
20	RAKE-like	bands	3.0	1

Step 4 - Preprocessing Comparison

document	method	tokens	unique	top_terms
Document 1	NLTK raw/lowercase	74	47	the, was, bolter, bolts, he, these, of, ee, roof, bundle, straps, on
Document 1	Stopword removal	37	28	bolter, bolts, roof, bundle, straps, loading, supplies, intake, four, foot, left, ont
Document 1	Porter stemming	37	28	bolter, bolts, roof, bundle, straps, loading, supplies, intake, four, foot, left, ont
Document 2	NLTK raw/lowercase	74	46	the, roof, was, bolts, of, ie, he, one, in, mouth, that, had
Document 2	Stopword removal	32	25	roof, bolts, one, mouth, preparing load, supply, tray, bolter, cut, bands, loose
Document 2	Porter stemming	32	25	roof, bolts, one, mouth, preparing load, supply, tray, bolter, cut, bands, loose

Step 5 - Related Terms under Broader Domain Concepts

broader_concept	document_1_related_terms	document_2_related_terms
roof-bolting work	roof bolts (1), bolts (4), roof bolter (1), bolter (4), intake roof bolter (1)	roof bolts (3), bolts (3), roof bolter (1), bolter (1)
material restraint	bands (1), strap (1), straps (2), bundle straps (1)	bands (1)
material handling	loading supplies (1)	load roof bolts (1)
roof-bolter components	-	supply tray (1)
mobile equipment components	-	scoop bucket (1), bucket (1)
unexpected material movement	strap struck (1)	struck him in the mouth (1)
oral injury	-	mouth (2), bridge (1), permanent tooth (1), tooth (1)
person reference	EE (2)	IE (2)

Step 7 - Exact or Contextual Equivalence

canonical	equivalent_forms	document_1_forms	document_2_forms
retaining band	band, bands, strap, straps, bundle straps	bands (1), strap (1), straps (2), bundle straps (1)	bands (1)
roof bolt	roof bolt, roof bolts, bolts	roof bolts (1), bolts (4)	roof bolts (3), bolts (3)
roof-bolting machine	bolter, roof bolter, intake roof bolter	bolter (4), roof bolter (1), intake roof bolter (1)	bolter (1), roof bolter (1)
employee	EE, employee	EE (2)	-
injured employee	IE, injured employee	-	IE (2)

Steps 5 and 7 - Interpretation

Exact or contextual equivalence is reported separately from broader domain relatedness. For example, band, bands, strap, and bundle straps refer to the same restraint concept in these narratives. Roof bolt and roof bolter are related to roof-bolting work but are not the same physical entity. Preserving that distinction is important for safety interpretation.

EE and IE are treated as person-reference abbreviations based on their contextual use. If the course expects stricter synonymy, these abbreviation groups can be removed or noted as context-dependent normalization rather than universal equivalence.

Step 6 - Dandelion - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	Dandelion	ee	Rank 1	2
2	Dandelion	bolts	Rank 2	4
3	Dandelion	bolter	Rank 3	4
4	Dandelion	strap	Rank 4	1

Step 6 - Dandelion - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	Dandelion	roof	Rank 1	4
2	Dandelion	bolts	Rank 2	3
3	Dandelion	bolter	Rank 3	1
4	Dandelion	scoop	Rank 4	1
5	Dandelion	bucket	Rank 5	1
6	Dandelion	permanent tooth	Rank 6	1
7	Dandelion	incident	Rank 7	1

Step 6 - Wordcount Phrase/Entity - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	Wordcount.com	intake roof bolter	Rank 1	1
2	Wordcount.com	four foot roof bolts	Rank 2	1
3	Wordcount.com	bundle straps	Rank 3	1
4	Wordcount.com	bolts on the bolter	Rank 4	0
5	Wordcount.com	strap laying loose	Rank 5	0
6	Wordcount.com	strap tension	Rank 6	0
7	Wordcount.com	ee was removing bolts	Rank 7	0

Step 6 - Wordcount Phrase/Entity - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	Wordcount.com	roof bolts	Rank 1	3
2	Wordcount.com	supply tray	Rank 2	1
3	Wordcount.com	roof bolter	Rank 3	1
4	Wordcount.com	ie (incident employee)	Rank 4	0
5	Wordcount.com	bridge in his mouth loosened	Rank 5	0
6	Wordcount.com	edge of the scoop bucket	Rank 6	1
7	Wordcount.com	permanent tooth	Rank 7	1
8	Wordcount.com	10/4/12	Rank 8	1

Step 6 - Domain Dictionary - Document 1

rank	method	term_or_entity	engine_output	literal_document_count
1	Domain dictionary	bolter	roof-bolting work	4
2	Domain dictionary	bolts	roof-bolting work	4
3	Domain dictionary	straps	material restraint	2
4	Domain dictionary	ee	person reference	2
5	Domain dictionary	intake roof bolter	roof-bolting work	1
6	Domain dictionary	loading supplies	material handling	1
7	Domain dictionary	bundle straps	material restraint	1
8	Domain dictionary	strap struck	unexpected material movement	1
9	Domain dictionary	roof bolter	roof-bolting work	1
10	Domain dictionary	roof bolts	roof-bolting work	1
11	Domain dictionary	bands	material restraint	1
12	Domain dictionary	strap	material restraint	1

Step 6 - Domain Dictionary - Document 2

rank	method	term_or_entity	engine_output	literal_document_count
1	Domain dictionary	roof bolts	roof-bolting work	3
2	Domain dictionary	bolts	roof-bolting work	3
3	Domain dictionary	mouth	oral injury	2
4	Domain dictionary	ie	person reference	2
5	Domain dictionary	struck him in the mouth	unexpected material movement	1
6	Domain dictionary	load roof bolts	material handling	1
7	Domain dictionary	permanent tooth	oral injury	1
8	Domain dictionary	scoop bucket	mobile equipment components	1
9	Domain dictionary	roof bolter	roof-bolting work	1
10	Domain dictionary	supply tray	roof-bolter components	1
11	Domain dictionary	bolter	roof-bolting work	1
12	Domain dictionary	bucket	mobile equipment components	1
13	Domain dictionary	bridge	oral injury	1
14	Domain dictionary	bands	material restraint	1
15	Domain dictionary	tooth	oral injury	1

Entity Extraction Metrics against Manual Entity Ground Truth

method	returned	ground_truth_hits	precision	recall	f1
Dandelion D1	4	2	0.500	0.133	0.211
Dandelion D2	7	1	0.143	0.067	0.091
Wordcount phrase/entity D1	7	3	0.429	0.200	0.273
Wordcount phrase/entity D2	8	4	0.500	0.267	0.348
Domain dictionary D1	12	7	0.583	0.467	0.519
Domain dictionary D2	15	9	0.600	0.600	0.600

Final Findings and Discussion

1. The two documents share a material-handling topic but use different surface wording.
   Phrase-level comparison is therefore more meaningful than single-word overlap alone.

2. FiveFilters returned useful material-handling terms but also generic fragments such as
   AS, TOP, and TEND. Those outputs increase coverage but reduce interpretability.

3. Wordcount.com preserved longer equipment, activity, hazard, and consequence phrases.
   Its results are generally easier to connect to the event sequence, although several
   outputs are full clauses rather than conventional named entities.

4. Dandelion returned only short labels. That limited output is itself informative because
   conventional named-entity tools are not optimized for short mining incident narratives.

5. Stopword removal reduces noise. Porter stemming merges some morphological variants but
   can reduce readability, so canonical concept mapping is more interpretable for this domain.

6. Online ranks, literal counts, TF-IDF weights, and RAKE-like scores are different measures.
   They should be compared by coverage and usefulness, not as directly equivalent numbers.
7. Highest exact-match term F1 in this run: FiveFilters D1 (precision=0.133, recall=0.222, F1=0.167).
8. Highest exact-match entity F1 in this run: Domain dictionary D2 (precision=0.600, recall=0.600,
F1=0.600).

Exact-match metrics are conservative. For example, 'bolts' and 'roof bolts' may refer to
the same material in context but will not match unless normalization is applied. The concept
tables should therefore be read alongside precision, recall, and F1.